TEXORA · LECTURE SHEET

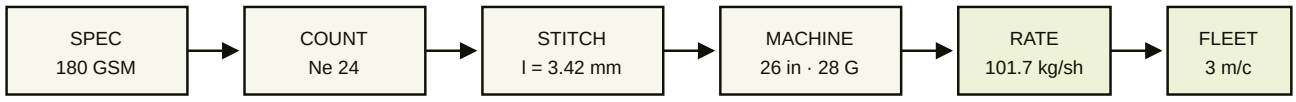
Overall Knitting Plan for a Particular Fabric

The knitting counterpart of Sheet 6: one fabric, one order, and every decision derived in sequence — count from GSM, stitch length from tightness, machine from width, rate from the machine, fleet from the rate, yarn from the order. A complete plan, with each step checking the one before.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 12 · SPECIALIZATION — FABRIC MANUFACTURING

1. The specification

ITEM	VALUE
Fabric	100% cotton single jersey, fully relaxed (finished) state
Target GSM	180 g/m²
Open width	176 cm
Order	12,000 kg, delivery in 15 days at 3 shifts/day (45 shifts)



The plan chain: no step can be taken before the one before it, and each is re-checked by the ones after.

Figure 1 — The six decisions of the plan, in the only order they can be made.

2. Steps 1–2: Count and stitch length

WORKED CALCULATION — FROM GSM TO YARN

By Sheet 9: tex = GSM ÷ (0.56 × WPC). The design is placed on a 28-gauge machine giving relaxed WPC 13/cm:

tex = 180 ÷ (0.56 × 13) = 24.7 → **Ne 24 (24.6 tex)**, delivering GSM = 0.56 × 13 × 24.6 = **179 g/m²**.

Tightness K = 1.45 (firm end of normal): l = √24.6 ÷ 1.45 = **3.42 mm**; CPC = 56 ÷ 3.42 = **16.4 per cm**.

3. Step 3: The machine from the width

WORKED CALCULATION — DIAMETER FROM NEEDLES

Needles required = width (cm) × WPC = 176 × 13 = 2,288 needles.
At 28 gauge, diameter = needles ÷ (π × gauge) = 2,288 ÷ (π × 28) = 26.0 in → **26-inch, 28-gauge machine**, whose exact needle count is π × 26 × 28 = **2,287** — WPC realises as 2,287 ÷ 176 = 13.0 ✓. The specification, the count and the machine agree; a plan whose three opening numbers do not agree is revised here, not at the knitting stage.

4. Steps 4–5: Rate and fleet

WORKED CALCULATION — PRODUCTION AND MACHINES

Machine fitted with 72 feeders, run at 18 rpm, efficiency 85%.
Courses/min = 18 × 72 = **1,296**; yarn per course = 2,287 × 3.42 ÷ 1,000 = **7.82 m**.
Consumption = 1,296 × 7.82 = 10,135 m/min → at 100%: 10,135 × 60 × 24.6 ÷ 10^6 = **14.96 kg/h**; at 85% = 12.72 kg/h → **101.7 kg per shift**.
Per machine over the order window: 101.7 × 45 shifts = **4,578 kg**.
Machines = 12,000 ÷ 4,578 = 2.62 → **3 machines** (capacity 13,734 kg against 12,000 kg ordered — a 14% margin, the tightest in the plan and therefore the plan's stated risk).

5. Step 6: Yarn, cloth and the closing checks

- **Yarn to purchase:** 12,000 × 1.02 (Sheet 10's 2% allowance) = **12,240 kg** of Ne 24.
- **Cloth quantity:** 12,000 ÷ 0.1791 = **67,002 m²** = 67,002 ÷ 1.76 = **38,069 m** — the finishing department's figure (and Sheet 10, Practice 3, checks it).
- **Structure check:** GSM by identity = WPC × CPC × l × tex ÷ 100 = 13 × 16.4 × 3.42 × 24.6 ÷ 100 = 179 g/m² ✓ — the plan's last act is to re-derive its first number by a second route.

Glossary

Overall plan — the complete chained calculation for one fabric: spec → count → stitch → machine → rate → fleet → yarn. **Margin** — installed capacity over demand, %; the plan's risk statement. **Realised WPC** — the width density the actual needle count gives at the specified width.

Key points

• Order of decisions is fixed: GSM chooses count (Ne 24), K chooses stitch length (3.42 mm), width + WPC choose the machine (26 in, 28 G, 2,287 needles).
• Rate by Sheet 8's chain: 101.7 kg/shift; fleet = order ÷ (rate × shifts) = 3 machines at 14% margin.
• Yarn = order × 1.02 = 12,240 kg; cloth = 67,002 m² = 38,069 m.
• Close the plan by re-deriving GSM from the chosen structure — agreement is the plan's signature.

Practice

1. The buyer widens the fabric to 190 cm at the same GSM and WPC. What diameter machine is needed at 28 gauge?
2. Only 2 machines are available for the order. How many shifts must each work, and what delivery (at 3 shifts/day) should be quoted?
3. Where is this plan most fragile, and what single measurement during the first shift would test it?

Answer key

1. Needles = 190 × 13 = 2,470; diameter = 2,470 ÷ (π × 28) = 28.1 in → **28-inch machine** (2,463 needles; WPC realises at 2,463 ÷ 190 = 13.0 ✓).
2. Shifts per machine = 12,000 ÷ (2 × 101.7) = 59 shifts → 59 ÷ 3 = 19.7 → quote **20 days**, not 15. The plan, not the buyer, absorbs the arithmetic.
3. The 14% fleet margin rests on the 85% efficiency inside the 101.7 kg rate. Measure the first shift's actual kg and stoppage record: if the realised rate is below 96 kg/shift, the 45-shift window no longer holds 12,000 kg even with perfect running thereafter, and the delivery conversation must happen on day 1, not day 14.